Complete Guide to SQL for Data Engineering: from Beginner to Advanced

with Deepak Goyal

LEARNING

Final Project Handout

Project Overview

This final project guides you through the essential steps of designing, building, and managing a database for claims and policies. The rubric below outlines the key tasks and achievement criteria for successful completion.

Key Tasks and Achievement Criteria

1. Create Database Schema

- **Design schema structure:**
  Develop a logical schema with all relations clearly defined and connected according to project requirements.
- **Include relevant fields:**
  Ensure fields such as CustomerID, PolicyID, Claim, and PolicyType are included for completeness and data normalization.
- **Establish relationships:**
  Properly connect tables (for example, policies to policy types, claims to customers).

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2. Populate the Database

- **Insert sample data:**
  Populate the database with a variety of sample data types to support comprehensive analysis.
- **Ensure data accuracy:**
  Verify that all inserted data adheres to the schema, with no misplaced or conflicting entries.

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3. Analytical Query Writing

- **Write analytical queries:**
  Create queries to calculate the total number of claims per policy type, determine monthly claim frequency, and compute average claim amounts using analytical functions.

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4. Performance Optimization

- **Add indexes:**
  Identify and add indexes to columns frequently used in WHERE clauses to enhance query performance.

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5. Manage Roles and Permissions

- **Define user roles:**
  Create roles such as ClaimAnalyst and ClaimsManager with appropriate access and editing capabilities.
- **Assign permissions:**
  Assign permissions to ensure access control aligns with security objectives.

Success Criteria

To excel in this final project, ensure you:

- Design a normalized and well-connected schema.
- Populate the database with accurate and varied sample data.
- Demonstrate proficiency in writing analytical queries.
- Optimize performance through effective indexing.
- Implement robust role-based access controls.